3DM Homework Hints and Solutions
Graph Theory Chapter 5

5.2: We can suppose, without loss of generality, that G is maximal planar (why?). Now modify the proof of Theorem 31, using the fact that $\delta(G) \geq 3$.

5.3: First use the information you have about the degrees of the vertices and what you know about the size of a maximal planar graph to prove that such a graph must have order 6. Now consider a vertex v in such a graph. What does Theorem 30 say about the neighbourhood of v?

5.4: Use the ideas in the proof that $K_{3,3}$ is nonplanar.

5.5: Discussed in class.

5.6: Use the fact that embedding on the plane is the same as embedding on the sphere. Which face of a plane graph embedded on the sphere corresponds to the outer face when embedded on the plane?

5.7: Since each component of a plane graph is itself a connected plane graph, Euler's Formula applies directly to each component. Now suppose there are k components and count the total numbers of vertices, edges, and faces to get a formula. Of course, when $k = 1$, your formula should be the same as Euler's Formula.

5.8: It's helpful in this context to introduce the idea of a *maximal outerplanar graph*. What would be a definition of such a graph? What would such a graph look like? For a, use the result from Question 5.6. For the formula in b, prove that a maximal outerplanar graph would have $m = 2n - 3$ by asking what its faces look like and then modifying the proof from Theorem 27.

5.11: Discussed in class.

5.14: Use Theorem 30.

5.15: You proved in a previous homework problem that the hypercubes are bipartite. You also calculated, in a (different) previous homework question, the order and size of the hypercube Q_t as a function of t. Use these and the ideas from the proof that $K_{3,3}$ is nonplanar (equivalently, your answer to Question 5.4) to determine for which values of t the graph Q_t is definitely nonplanar. It should then be relatively easy to prove that for the other values of t this graph is planar.